ตัวอย่าง · ตัดข้อมูลลูกค้าและราคาออกแล้ว · Sample: customer details and prices removed

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PROPOSAL

29.25 kWp Off-Grid Hybrid Solar System — Mounting Structure Options

Reference: QO-20261015-01 Date: 15 October 2026
Prepared by: Solar ESS Co. Ltd. Price validity: to be advised

1. Introduction

Thank you so much for thinking of me for your project. I've put together a complete off-grid solar system for you — 29.25 kWp of panels with 60 kWh of battery — so you can enjoy clean, quiet power day and night without relying on the grid. To keep things simple, I'm giving you two choices for the structure under the panels. Everything else — the equipment, my team's work, the services and the warranties — is exactly the same either way. Option A is my new A-shape frame with adjustable tilt, sitting on concrete ballast blocks; it stands higher, looks neat and gets a bit more out of the bifacial panels. Option B is the low ground-mounted structure I've used before — proven, simple and the more economical one.

My suggestion is to start from the plan you would really love to have. That way you can see clearly how far the budget can go, and from there you and I can trim it step by step and keep what is truly significant to the success of your project. To be open with you from the start: Option A will not come out cheaper than Option B. It costs about THB – more including VAT, and all of that difference is in the structure — the A-shape frames, the ballast blocks and the crane. So the choice is really about how you want the array to look and perform. And once the plan is settled, I'm happy to talk about a discount with you as the final step.

How I suggest moving forward

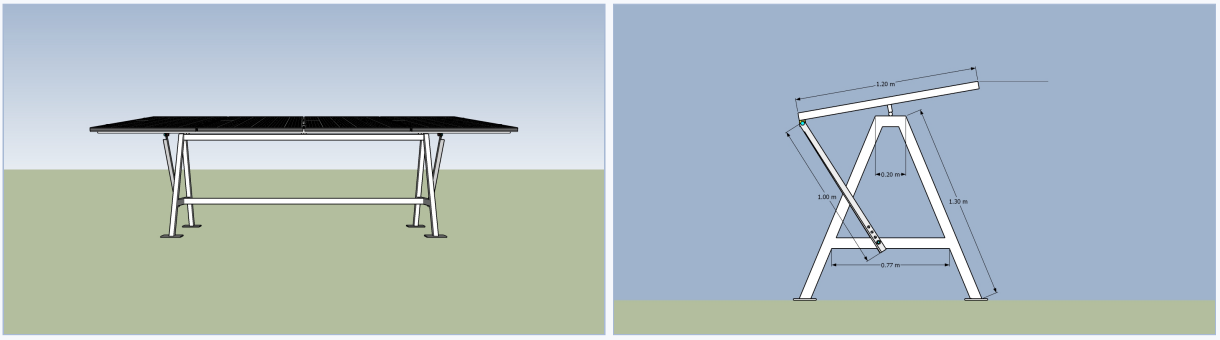
1. **Start from your ideal plan** — the system and structure you would really love to have, as set out in this proposal.
2. **Trim and prioritise together** — you and I go through it item by item and keep what truly matters to the success of the project, so you can see how far the budget can go.
3. **Settle the plan** — agree on the option and the final scope.
4. **Final discussion and negotiation** — this is the point where I look at which part of my work and price I can give back to meet your budget, including any discount.
5. **Decide together** — if it works for you and for me, I prepare the contract and the project moves ahead. If it does not, you and I close the process there, as friends, with sincere appreciation for each other's time and contribution.

2. Solar system — common to both options (Part 1)

- **Inverter** — 1 × Solis S6-EH3P50K-H hybrid inverter, 50 kW, 3-phase, with automatic generator control
- **Solar panels** — 45 × LONGi Hi-MO X10 650 W bifacial (HPBC 2.0) = 29.25 kWp, three strings of 15
- **Battery** — Solis FlexCore-ID LiFePO4, 3 × 20 kWh = 60 kWh, with CB250A control box; expandable to about 100 kWh
- **Electrical** — MDB with MCCBs, 100 A ATS for a future generator, AC/DC surge protection, DC combiner with string fuses and breakers, 4 mm² PV cable with matched MC4-EVO2 connectors, Thai Yazaki AC cable, earthing and ground test box
- **Services** — installation, testing and commissioning with an off-grid start-up demonstration; handover documents, single line diagram and as-built drawing; training; panel cleaning and electrical safety audit once a year for 5 years
- **Warranty** — 5 years installation workmanship; inverter 10 years, battery 10 years, panels 15-year product / 30-year performance
- **Contingency** — 5% on the main equipment (inverter, battery and panels) to cover market price movements

Part 1 price (equipment, materials and installation): **THB –** excl. VAT. The itemised bill of materials for Part 1 and for each structure option is given in the Appendix.

3. Option A — A-shape structure on ballast blocks



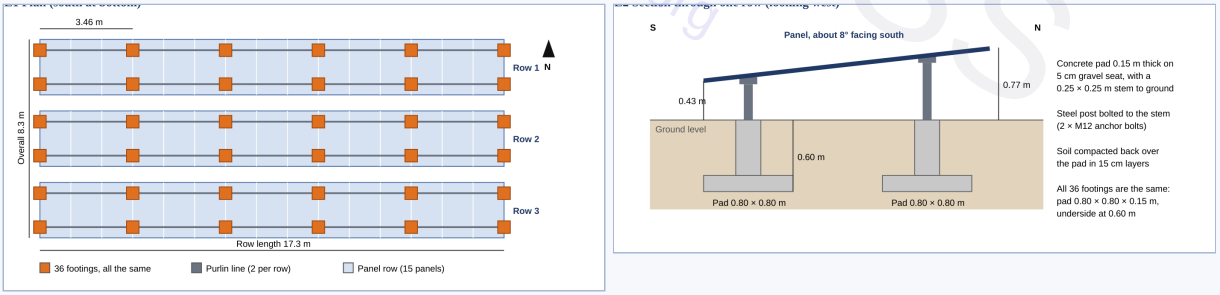
Typical section, front view (left) and side view of one A-frame (right)

- 45 panels in landscape in **three rows of fifteen**, each row about 36.0 m long; rows about 2.0 m apart; array area about 36 × 5.3 m
- **42 adjustable-tilt A-frames** (14 per row, about 2.68 m apart), steel tube 2"×2" × 2.0 mm with 1.20 m rafters and adjustable 1.00 m struts; tilt about 10°, adjustable; panels about 1.2–1.45 m above the ground
- Two continuous 4"×2" × 2.0 mm purlins per row; panels bolted at their mounting holes; zinc-rich primer and PU top coat
- **84 precast reinforced concrete ballast blocks**, 0.50 × 0.50 × 0.35 m (about 210 kg), set on the ground on compacted gravel — no excavation; designed to DPT 1311-50 for Chiang Rai with a safety factor of about 1.56 against uplift
- Crane on site for two days to place the blocks and lift the structure
- Raised panels improve the rear-side (bifacial) yield and keep the panels clear of splash and low growth

Structure price (STR-02, materials, labour and crane, incl. 5% allowance): **THB –** excl. VAT.

Option A total: THB – including VAT

4. Option B — Low ground-mounted structure (lower cost)



Plan (left) and section through one row (right)

- 45 panels in portrait in **three rows of fifteen**, each row one string; array about 17.3 × 8.3 m with 0.6 m walkways
- Low steel structure at a single slope of about 8°, panels about 0.43–0.77 m above the ground; two 4"×2" × 2.0 mm purlin lines per row on 36 supports; panels bolted directly to the purlins; zinc-rich primer and PU top coat
- **36 reinforced concrete pad footings** designed for the Chiang Rai wind uplift (DPT 1311-50) with a safety factor of at least 1.5; the footing type may be adapted on site where ground conditions call for it, within the turnkey price
- Same solar system, services and warranties as Option A

Structure price (STR-02, materials and labour, incl. 5% allowance): **THB –** excl. VAT.

Option B total: THB – including VAT

5. Price comparison

THB	Option A — A-shape + ballast	Option B — low ground mount
Part 1 — Solar system (equipment, materials, installation)	–	–
Part 2 — Mounting structure and footings (STR-02)	–	–
Subtotal (excl. VAT)	**–**	**–**
VAT 7%	–	–
Total (incl. VAT)	**–**	**–**

Payment in three installments

Installment	Option A	Option B
1 — 30% on signing of the contract	–	–
2 — 40% on delivery of the main equipment to the site	–	–
3 — 30% on completion, testing, commissioning and acceptance	–	–

6. Your price, secured

Just so you know, solar equipment prices here move with the market and can change from day to day. I checked every price in this proposal against the market on 15 October 2026, and I'll let you know how long they hold. Once you sign and I receive the first installment, your price is locked — it won't change for any reason, whatever happens to equipment or transport costs, unless you ask me for extra work in writing.

7. A few things to keep in mind on timing

There's no rush from my side — the timing is completely up to you. I just want you to have the full picture, so here's how timing can work in your favour:

- **My team** — my most experienced leader and my best installers are free right now. I schedule my teams in the order projects are confirmed, so if you confirm early, the same people can stay with your project from the first footing to the switch-on.
- **Your price** — prices follow the market, but once you sign and I receive the first installment, your price is locked and market changes no longer affect you.
- **Your equipment** — as soon as you confirm, I order your inverter, battery and panels and keep them aside for you.
- **The season** — the cool, dry months are honestly the best time to build; the work goes faster and the site is easier.
- **Batteries next year** — China's export tax rebate on batteries is due to end on 1 January 2027, so battery prices may go up next year.
- **Your savings** — the sooner your system is running, the sooner you start enjoying your own free, clean electricity.

8. Terms common to both options

- Turnkey contract: Solar ESS is the single responsible contractor for the solar system, its mounting structure and footings
- Completion within 45 working days (rain days and days off not counted) from signing, receipt of the first installment and site readiness
- Floor and ground works (cement floors, crushed stone, pebbles, weed membrane, grading) are not included
- Generator and its installation are by the generator supplier; Solar ESS recommends the specification and connects it at the ATS
- VAT 7% is charged on the whole price, materials and labour

9. Next step

Whenever you're ready, just tell me which one you like — A or B — and I'll confirm the final price and send you the contract to look over. If you'd like to talk it through, I'm always happy to come by the site, answer any question, walk you through the design or talk about a discount over a coffee. Thanks again for considering me and my team — I'd really love to build this with you.

Warm regards,

Kris Dhirad ityakul
Solar ESS Co. Ltd.

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Appendix — Bill of Materials and Prices

Prices in THB excluding VAT. Unit prices follow current Thai market prices and were checked against the market on 15 October 2026; the final price is confirmed on the date of signing (Section 6).

A1 Part 1 — Solar system (common to both options)

#	Code	Item	Product	Qty	Amount
1	INV-01	Hybrid inverter 50 kW, 3-phase	Solis S6-EH3P50K-H	1 unit	–
2	BAT-01	Battery pack 20 kWh, 10-yr warranty	Solis FlexCore-ID-BAT20kWh (10Y)	3 pack	–
3	BAT-01	Battery control box 250 A	Solis FlexCore-ID-CB250A	1 unit	–
4	BAT-01	Battery accessory pack	Solis FlexCore-ID-BAT20kWh-AP2	1 set	–
5	PV-01	Solar panel 650 W bifacial, incl. transport	LONGi Hi-MO X10 650 W	45 pc	–
6	CONT	Contingency 5% on main equipment (inverter, battery and panels, items 1–5)	5% × 656,280	1 lot	–
7	ELE-01	MDB enclosure	KJL steel cabinet	1 set	–
8	ELE-01	Main incoming MCCB 100 A	CHINT NXM-125S/3P	1 pc	–
9	ELE-01	Inverter grid-port MCCB 100 A	CHINT NXM-125S/3P	1 pc	–
10	ELE-01	Inverter backup-output MCCB 100 A	CHINT NXM-125S/3P	1 pc	–
11	ELE-01	ATS 4P 100 A	SUNTREE / CSQ / Geya	1 set	–
12	ELE-01	Outgoing feeder breakers	CHINT NXB-63 / NXM	1 lot	–
13	ELE-01	AC surge protection Type II	SUNTREE SUP1H-40 + SCB8-63	1 set	–
14	ELE-01	Busbars and panel accessories	Cu busbars, N/PE bars, DIN, labels	1 lot	–
15	ELE-02	DC combiner enclosure	KJL outdoor, 3 in / 3 out	1 pc	–
16	ELE-02	DC fuse holder + gPV fuse 32 A	SUNTREE SRD-30 + SRF-30	6 set	–
17	ELE-02	DC breaker 1000 V 32 A	SUNTREE SL7-63 4P	3 pc	–
18	ELE-02	DC surge protection Type II	SUNTREE SUP2H1-PV 3P	3 pc	–
19	ELE-02	Combiner accessories	DIN rail, terminals, glands	1 lot	–
20	ELE-03	PV cable 4 mm², red	LINK H1Z2Z2-K	50 m	–
21	ELE-03	PV cable 4 mm², black	LINK H1Z2Z2-K	480 m	–
22	ELE-03	MC4 connectors, genuine, matched to panel leads	Stäubli MC4-EVO2	6 pair	–
23	ELE-03	Module and structure bonding cable	Thai Yazaki 6 mm² G/Y	1 lot	–
24	ELE-03	DC conduit and fixings	UV-resistant conduit	1 lot	–
25	ELE-04	AC power cable 35 mm²	Thai Yazaki THW	100 m	–
26	ELE-04	Earth cable 16 mm²	Thai Yazaki THW G/Y	50 m	–
27	ELE-04	AC conduit / trunking and lugs	Conduit, glands, lugs	1 lot	–
28	–	Ground test box	KUMWELL GYPTB	1 pc	–
29	SVC-01	Installation, testing, commissioning	29,250 W × – THB/W	1 lot	–
30	SVC-02	PV cleaning + electrical safety audit, yearly	5 years × 5,000	5 year	–
Subtotal Part 1					**–**

A2 Part 2, Option A — A-shape structure on ballast blocks (STR-02)

#	Item	Specification	Qty	Unit	Amount
30	Purlins			–	–

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		Tube 4"×2" × 2.0 mm, 6 m, on edge; 2 continuous lines per row (36.0 m), spliced	38 length		
31	Rafters	Tube 4"×2" × 2.0 mm, 6 m; 42 rafters of 1.20 m	9 length	–	–
32	A-frames and beams	Tube 2"×2" × 2.0 mm, 6 m; legs, cross bars, apex caps, struts, top and tie beams in every span	73 length	–	–
33	Strut adjusters	Tube 1.5"×1.5" × 2.0 mm, 6 m; 0.5 m sliding sleeve per strut	4 length	–	–
34	Plates	Base plates 200×200×6 mm (84), pivot brackets, purlin splice and connection plates	305 kg	–	–
35	Anchor bolts	M12 J-bolt cast into each ballast block + nuts/washers; 2 per block	168 set	–	–
36	Pivot and tilt bolts	M12 bolts with lock nuts and washers; 3 per A-frame + spare	136 set	–	–
37	Purlin bolts	M12 bolts, nuts, washers; purlin to rafter and purlin splices	312 set	–	–
38	Panel fixing	M8 rivet nut + M8 × 45 A2-70 bolt; 4 per panel + 8 spare	188 set	–	–
39	Welding consumables	Electrodes, grinding and cutting discs	1 lot	–	–
40	Paint	Zinc-rich primer + PU top coat (about 190 m²)	1 lot	–	–
41	Ballast concrete	240 ksc: 84 precast ballast blocks 0.50 × 0.50 × 0.35 m (7.35 m³ + waste)	7.8 m³	–	–
42	DB10	DB10 SD40, 10 m; reinforcing mesh in each block	35 bar	–	–
43	Binding wire	Tie wire #18	5 kg	–	–
44	Block moulds	Steel/plywood moulds, reused	1 lot	–	–
45	Block bed	Gravel, 5 cm compacted under each block only (not ground cover)	1.6 m³	–	–
46	Block transport	Truck from the Contractor's premises to the Site (84 blocks, about 18 t)	3 trip	–	–
47	Crane	Crane on site for placing the blocks and lifting the structure, 2 days × 10,000	2 day	–	–
48	Earthing	3 × earth rod 5/8" × 2.4 m, bonding cable and clamps along the three rows	1 lot	–	–
49	Structure labour	Casting 84 ballast blocks (10 man-days); setting out, gravel bed and placing the blocks (6); fabricating 42 A-frames, beams and purlins (23); painting (10); erecting and aligning the three rows, earthing (10)	59 man-day	–	–
Items (materials, crane and labour)					–
Allowance 5%					–
Subtotal Part 2 — Option A					**–**

A3 Part 2, Option B — Low ground-mounted structure (STR-02)

#	Item	Specification	Qty	Unit	Amount
30	Purlins	Tube 4"×2" × 2.0 mm, 6 m, on edge; 2 lines per row	18 length	–	–
31	Posts	Tube 2"×2" × 2.0 mm, 6 m; 18 posts of 0.40 m + 18 posts of 0.60 m	3 length	–	–
32	Plates	Base, cap and purlin splice plates	86 kg	–	–
33	Anchor bolts	M12 J-bolt L = 300 mm + nuts/washers; 2 per footing	72 set	–	–
34	Purlin bolts	M12 bolts, nuts, washers	120 set	–	–
35	Panel fixing	M8 rivet nut + M8 × 45 A2-70 bolt; 4 per panel + 8 spare	188 set	–	–
36	Welding consumables	Electrodes, grinding and cutting discs	1 lot	–	–
37	Paint	Zinc-rich primer + PU top coat	1 lot	–	–
38	Concrete	240 ksc: 36 pads 0.80 × 0.80 × 0.15 m + 36 stems 0.25 × 0.25 × 0.45 m (4.5 m³ + waste)	4.9 m³	–	–
39	DB10	DB10 SD40, 10 m; pad mesh + 4 bars per stem	36 bar	–	–
40	Binding wire	Tie wire #18	3 kg	–	–
41	Formwork	Stem forms, reused	1 lot	–	–
42	Footing seat	Gravel, 5 cm compacted under each footing pad only	1.3 m³	–	–

43	Plate compactor	Hire for backfill compaction in 15 cm layers	2 day	–	–
44	Earthing, row 1	2 × earth rod 5/8" × 2.4 m, bonding cable and clamps	1 lot	–	–
45	Earthing, rows 2 and 3	Bonding to the row 1 earth system	2 lot	–	–
46	Structure labour	Digging 36 footings, casting and compacted backfill (14 man-days); fabrication and erection of the three rows (20)	34 man-day	–	–
Items (materials and labour)					–
Allowance 5%					–
Subtotal Part 2 — Option B					**–**

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